VNR Vignana Jyothi Institute of Engineering

Mechanical Engineering | Course Code: ME201

Mechanics of Solids & Strength of Materials - Laboratory Manual & Practi

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Syllabus Focus: ME201 Laboratory Practical Manual & Viva Voce Handbook

Summary: Structured laboratory workbook detailing experiment objectives, baseline toolchains, e

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Simple Stresses, Strains & Elastic Moduli

" Syllabus Scope: Normal and shear stress, Hookes law, stress-strain curves for ductile and brittle mater

Unit 2: Compound Stresses & 2D Mohr Circle

" Syllabus Scope: Stresses on inclined planes, principal stresses, principal planes, maximum shear stress

ME201: Mechanics of Solids & Strength of Materials - Advanced Topics & Assessment

Academic Track: Mechanical Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Shear Force & Bending Moment in Beams

" Beam support classifications, point loads, uniformly distributed loads UDL, uniformly varying loads UVL, SFD

Unit 4: Flexural Stresses, Shear Stresses & Deflection

" Pure bending theory Euler-Bernoulli, section modulus, shear stress distribution across I and T sections, dou

Unit 5: Torsion of Circular Shafts & Column Buckling

" Torsion formula for solid and hollow shafts, polar modulus, angle of twist, Eulers column buckling theory, R

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

Lab 1: Tensile Strength and Ductility Determination of Mild Steel Specimen on Universal Testing Machine U

Lab 2: Torsion Testing of Solid Steel Shaft to Measure Modulus of Rigidity G

Lab 3: Deflection Testing of Simply Supported and Cantilever Beams to Verify Flexural Rigidity EI

Lab 4: Brinell and Rockwell Hardness Characterization of Ferrous and Non-Ferrous Alloys

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* Ferdinand P. Beer, E. Russell Johnston - Mechanics of Materials McGraw-Hill 7th Ed

* James M. Gere, Barry J. Goodno - Mechanics of Materials Cengage Learning 8th Ed

Authorized NoteVault Educational Document " Verified for Registered University Students